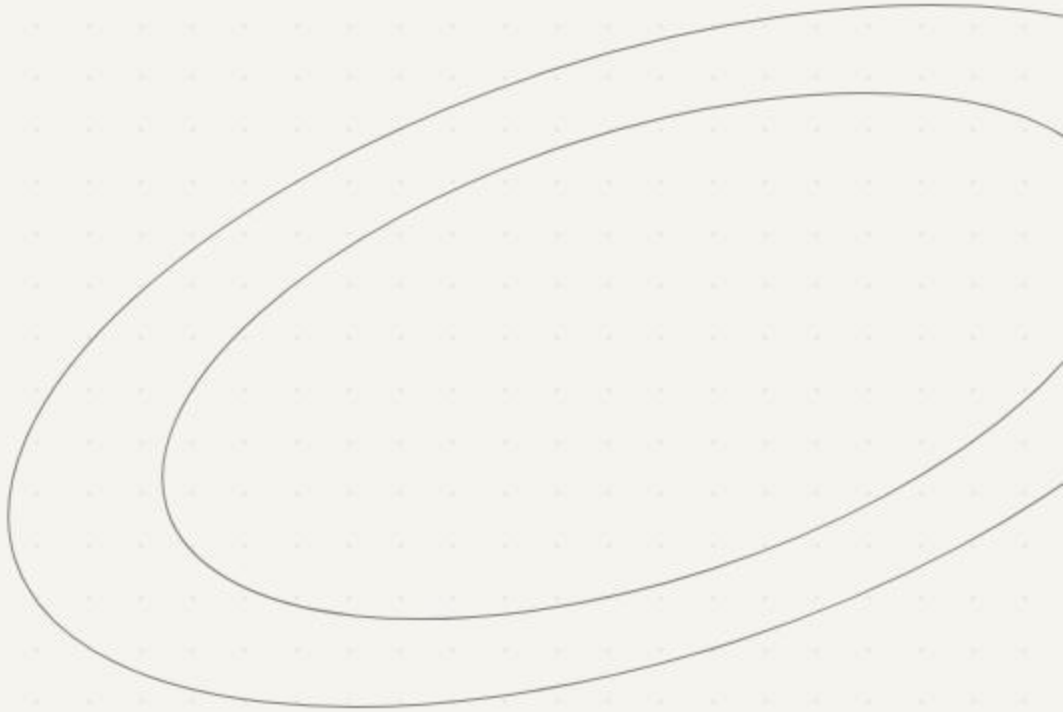
CELESTIAL MECHANICS × AI · 2026

From the next token to the next theorem

How a model works — and how I use it in practice



Bowen Liu

Overview: from token to theorem

Click a section card to jump; click a tag for a key page.

00 · OPENING + TOKEN

Token

Ten integers · vocabulary · billing

p. 3–8

Ten integers | Vocabulary

01 · MODEL

One function

Distribution

p. 9

Distribution

02 · TRAINING & PROBLEM SOLVING

Learning from reward

RL loop · math and code · GRPO

p. 10–23

RL loop | GRPO formula | REFLEX | Stateless

03 · SYSTEMS & ENGINEERING

Put the model in a system

Context · agents · code review

p. 24–35

Claude Code | System map

04 · AI × MATH

From tools to search

Acceleration · timeline · pattern

p. 36–41

Timeline | Pattern

05 · CONCLUSION

Back to practice

Workflow · compounding · close

p. 42–48

Two attempts | Close

6 sections · 48 pages · Click any card to start; `N` notes, `F` fullscreen.

The GRPO objective

DeepSeekMath: group-relative advantage with clipping and a KL constraint.

$$J_{\mathrm{GRPO}}\left(\theta\right) = E_{q \sim P(Q),\, \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\mathrm{old}}}} \left[\frac{1}{G} \sum_{i=1}^{G} \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} L_{i,t} \right]$$

PER-TOKEN TERM

$$L_{i,t} = \min\left[\rho_{i,t}\left(\theta\right) \hat{A}_{i,t},\, \mathrm{clip}\left(\rho_{i,t}\left(\theta\right), 1-\varepsilon, 1+\varepsilon \right) \hat{A}_{i,t} \right] - \beta D_{\mathrm{KL}}\left[\pi_\theta \| \pi_{\mathrm{ref}} \right]$$

PROBABILITY RATIO

$$\rho_{i,t}\left(\theta\right) = \frac{\pi_\theta\left(o_{i,t} \mid q, o_{i,<t}\right)}{\pi_{\theta_{\mathrm{old}}}\left(o_{i,t} \mid q, o_{i,<t}\right)}$$

GROUP-RELATIVE ADVANTAGE

$$\hat{A}_{i,t} = \frac{r_i - \mathrm{mean}\left(r_1, \ldots, r_G\right)}{\mathrm{std}\left(r_1, \ldots, r_G\right)}$$

Raise high-advantage responses while **clip and KL** limit policy drift.

Source: Shao et al., *DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models*, arXiv:2402.03300, 2024.

REFLEX: search engine in the RL loop

E-commerce query expansion: generate → retrieve → reward result quality.

Query + top-k products → LLM expands → Search engine → Union of results → Reward ↺

REWARD · NO HUMAN LABELS

Relevance Rel(Y)/Rel(q) + coverage Ret(Y)/Ret(q); format violations score 0.

TRAINING

SFT warm start on 10k → GRPO on 100k real queries; 8×H200.

QUERIES WITH POSITIVE GAIN · %

SETTING	ΔCOV	ΔREL
SFT ONLY	12.8	8.9
GRPO ONLY	57.5	25.2
WARM START + GRPO	72.8	32.4

10,000 queries · 5 samples · LLaMA 3.2-3B

> GRPO turns **better retrieval** into an optimizable reward.

Zhang, Liu, et al., REFLEX, WSDM 2026 · ablation on this dataset and backbone.

The model is stateless

CALL A

```
system    You are a celestial mechanics assistant
user      What is the Hill region?
assistant The Hill region is …
user      Why does it matter?
assistant Because it defines …
```

CALL B · A FRESH CALL

```
system    You are a celestial mechanics assistant
user      Why does it matter?
assistant ("it" has no referent here)
```

> The model is stateless; a "conversation" is the client **resending** the history.

Parameters stay fixed at inference time; state lives only in the context.

After training: four limits, three system capabilities

The model predicts; state, computation and reliability live outside it.

01 · MEMORY & CONTEXT

No memory, limited window

- Parameters fixed at inference; knowledge in weights
- A "conversation" resends history
- 4k → 1M, but long inputs still lose the middle

→ Context management · retrieval · memory

02 · EXACT COMPUTATION

Generated, not executed

- Arithmetic is token prediction, not calculation
- Plausible answers can still be wrong
- Errors compound over long tasks

→ Function calling · call a program

03 · PROBABILISTIC OUTPUT

Same input, different output

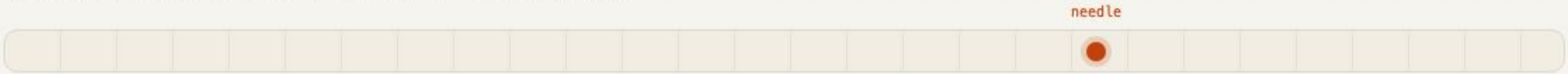
- Output is sampled, not a deterministic function
- Errors can sound confident and plausible
- Long runs need stopping and rollback

→ Harness · verification · permissions · evals

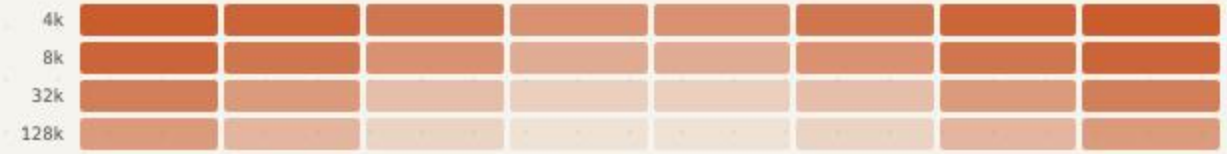
> Four limits, three system capabilities: **state, computation, reliability**.

Context is finite

Needle in a haystack: hide one sentence in a long document



Retrieval accuracy (illustrative)



Horizontal axis: needle position 0 → 100%

METHOD

Hide a sentence, vary length and position, ask what it was

OBSERVED FAILURE MODE

Length and position can affect retrieval; behavior varies by model and task

IMPLICATION

A big window is a budget; retrieval still matters

Schematic, not measured data · See Liu et al., Lost in the Middle (2023).

> Window capacity and **effective use of context** are different measures.

When code outruns human review

Claude Code's review problem: code scales faster than human attention.

SCALE PRESSURE · ANTHROPIC MONOREPO

2,500 → 30,000+

Changes landing per day · May → Sep 2026

Claude Code has a separate repository.

PRODUCTION

Agents write more code, faster.

BOTTLENECK

Human review time stays limited.

CONSEQUENCE

Line-by-line review cannot keep up.

Goal: reduce **review + repair time**, with reverts and incidents as guardrails.

> The next gain comes from **redesigning review**.

Source: Osmani, Lunt & Wang · 15 Oct 2026 · How the Claude Code team does code review · Team-reported practice.

Put the model back in the system



Inner ring: one call · Middle ring: across calls · Outer ring: engineering boundary

01	**LLM**	A probability generator: tokens in, a distribution over the next token out
02	**Prompt**	This call's conditions: system / user / few-shot
03	**Context**	Every token visible to this call; the window is finite
04	**Memory**	State beyond the window: write / retrieve / compress / forget
05	**Agent loop**	Observe → act → observe; state lives outside
06	**Harness**	Tools, sandbox, permissions, verification, logs

> The model **proposes**; the system manages context and state, runs tools, and **checks results**.

From the next token to the next theorem

`token` → `probability` → `candidate` → `verification` → **`theorem`**

Probability gives breadth

Verification gives correctness

People set the direction

> The next theorem still begins with the **next token**.